SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1741**

CO1-AT2-Constraint-Based Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 20%

QUESTIONS:3 **Total Marks:30**

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that: Constraints:

   i. D1 cannot work Night shift ii. D2 must work before D3 (Morning < Afternoon < Night) iii. Only one doctor per shift iv. D3 cannot work Morning

   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths. Constraints:

   i. Movement allowed: Up, Down, Left, Right ii. Cost of each move = 1

   iii. Cannot pass through obstacles

   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.
   The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

   Constraints:

   - The robot can move up, down, left, right

- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved. c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

c) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Question 1: Backtracking Search for Doctor Shift Assignment

Introduction

Backtracking search is a constraint satisfaction problem (CSP) technique that assigns values to variables one by one. If an assignment violates any constraint, the algorithm goes back and tries another value.

Given constraints

- Doctors: D1, D2, D3
- Shifts: Morning (M), Afternoon (A), Night (N)
- D1 cannot work Night.
- D2 must work before D3.
- Only one doctor per shift.
- D3 cannot work Morning.
- Every doctor must be assigned exactly one shift.

Step-by-step solution

Step 1: Assign D1

Possible shifts for D1 are Morning and Afternoon because D1 cannot work Night.

Choose:

\[D1 = M \]

Constraint check: Valid.

Step 2: Assign D2

Morning is already occupied by D1. D2 can take Afternoon or Night.

Choose:

\[D2 = A \]

Constraint check: Valid so far.

Step 3: Assign D3

Morning and Afternoon are occupied. Therefore, the only available shift is Night.

Choose:

\[D3 = N \]

Constraint check: Valid because D3 cannot work Morning, and D2 works before D3.

Backtracking and constraint checks

Step	Assignment	Result
1	D1 = M	Valid
2	D2 = A	Valid
3	D3 = N	Valid
4	Check unique shifts	Satisfied
5	Check D2 before D3	A before N — satisfied
6	Check all doctors assigned	Satisfied

The first assignment sequence succeeds, so backtracking is not required.

Final valid schedule

Shift	Doctor
Morning	D1
Afternoon	D2
Night	D3

Conclusion: Backtracking search finds a valid assignment by checking each constraint after assigning a shift. The final schedule satisfies all the given constraints.

Question 2: Robot Navigation Using BFS and Manhattan Distance

Introduction

Breadth-First Search (BFS) explores nodes level by level. It uses a FIFO queue and finds the shortest path when every movement has the same cost.

Important: Manhattan distance is a heuristic used by algorithms such as A*. Standard BFS does not use a heuristic or a priority queue. Since the question specifies BFS, we will demonstrate BFS and calculate Manhattan distance for the nodes.

Given constraints

- Grid size: \(5 \times 5\)
- Start: S at (0,0)
- Goal: G at (4,4)
- Movement: Up, Down, Left, Right
- Cost per move: 1
- Obstacles cannot be crossed.

Since obstacle locations are not specified in the question, assume the following grid for demonstration.

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G

Assumed grid: ■ represents an obstacle; S is the start and G is the goal.

Manhattan distance heuristic

The Manhattan distance is calculated as:

\[h(n)=|x_n-x_G|+|y_n-y_G| \]

For the start position (0,0):

$$ h(S)=|0-4|+|0-4|=8 $$

Thus, at least 8 moves are needed to reach the goal.

BFS algorithm steps

1. Insert the start node S into the queue.
2. Remove the first node from the queue and expand its valid neighbours.
3. Ignore obstacles and already visited nodes.
4. Add newly discovered nodes to the end of the queue.
5. Continue until G is reached.
6. Trace the parent nodes to obtain the path.

Step-by-step queue updates

Use the neighbour order Right, Down, Left, Up.

Step	Expanded node	New valid nodes added	Queue after expansion
1	(0,0)	(0,1), (1,0)	(0,1), (1,0)
2	(0,1)	(0,2)	(1,0), (0,2)
3	(1,0)	(2,0)	(0,2), (2,0)
4	(0,2)	(0,3), (1,2)	(2,0), (0,3), (1,2)
5	(2,0)	(3,0)	(0,3), (1,2), (3,0)

The process continues by expanding nodes in FIFO order, avoiding the blocked cells. Eventually, the goal is reached.

Optimal path

One valid shortest path is:

$$ \begin{aligned} (0,0)&\rightarrow(0,1)\rightarrow(0,2)\\ &\rightarrow(0,3)\rightarrow(0,4)\\ &\rightarrow(1,4)\rightarrow(2,4)\\ &\rightarrow(3,4)\rightarrow(4,4) \end{aligned} $$

The robot moves four steps right and four steps down.

$$ \text{Total cost}=8\times1=\boxed{8} $$

Conclusion

BFS finds the optimal path because every move has equal cost. The Manhattan distance of 8 confirms the lower bound on path length, and the demonstrated route achieves it. If the problem specifically requires Manhattan distance to determine node-expansion priority, A* search should be used instead of BFS.

Question 3: Constraint-Based Problem Solving for an Autonomous Rescue Robot

Introduction

An autonomous rescue robot is an intelligent agent designed to locate survivors in a disaster-affected building. The building is represented as a grid containing free rooms, obstacles, and

risky zones. The robot has limited sensing capability and must find a safe, minimum-cost path to the survivor.

(a) Analysis of constraints

1. Movement constraints

The robot can move only Up, Down, Left, or Right. Diagonal movement is not allowed.

2. Obstacle constraints

Blocked rooms cannot be entered. The robot must identify alternative routes around obstacles.

3. Partial observability

The robot can observe only adjacent cells. It does not know the complete building layout initially and must update its knowledge as it moves.

4. Movement cost

Each normal movement costs 1 unit. Entering a risky zone adds 2 units, making the total cost of that move 3 units.

$$\text{Total cost}=\text{Normal movement cost}+\text{Risk cost}$$

5. Safety constraint

The robot should avoid dangerous areas whenever possible and choose a safer route when the reduction in risk justifies the additional travel cost.

6. Dynamic knowledge

The robot updates its internal map after detecting obstacles, free rooms, and risky zones. This allows it to change its route when new information becomes available.

(b) Solution using Uniform Cost Search and an Online Search Agent

Uniform Cost Search (UCS) expands the node with the lowest accumulated path cost. An online search agent uses observations from the environment to decide where to move next.

Steps involved:

1. Initialize: Place the robot at S and create an initial map using the information currently available.
2. Sense: Observe the adjacent cells to identify free paths, obstacles, and risky zones.
3. Generate successors: List all reachable neighbouring cells.
4. Calculate costs: Assign cost 1 to normal moves and cost 3 to moves entering risky zones.
5. Select the lowest-cost node: Use a priority queue ordered by accumulated path cost $g(n)$.

6. Move and update: Move to the selected location, update the map, and revise the search if new information is discovered.
7. Repeat: Continue sensing and planning until the survivor at G is reached.

The UCS evaluation function is:

$$ f(n)=g(n) $$

where $g(n)$ is the total cost from the starting position to node n.

Example of path-cost comparison

Suppose two possible routes are available:

Route	Normal moves	Risky-zone entries	Total cost
A	6	1	$6+3=9$
B	8	0	8

Route A costs 9 units, while Route B costs 8 units. Therefore, the robot selects Route B, which avoids the risky zone and has a lower total cost.

Note: This is an illustrative cost comparison, not a specific map supplied in the question. In an unknown building, UCS must account for newly discovered information as the robot explores.

(c) Application of AI concepts

1. Intelligent agents

An intelligent agent perceives its environment through sensors and acts through movement mechanisms.

- Sensors detect nearby obstacles and hazards.
- The internal model stores the discovered map.
- The decision-making system selects the next movement.
- The actuators move the robot toward the survivor.

A model-based, goal-based agent is appropriate because it maintains knowledge and works toward reaching G.

2. Rationality

A rational agent selects actions that maximize its expected performance based on available information.

The robot should:

- Reach the survivor at minimum cost.
- Avoid obstacles.
- Reduce exposure to risky zones.
- Update its decisions when new information becomes available.

Rationality does not mean having complete knowledge. It means selecting the best action based on the information currently available.

3. Environment type

Property	Classification
Observability	Partially observable
Determinism	Usually deterministic if movement outcomes are predictable
Structure	Sequential
Change over time	Dynamic if hazards or paths change
State space	Discrete
Knowledge	Initially unknown

These characteristics require the robot to sense, remember, plan, and adapt continuously.

(d) Justification for the chosen approach

A model-based online search agent combined with UCS is suitable because:

1. It handles an initially unknown environment by updating the map as the robot explores.
2. UCS considers different movement costs and finds a minimum-cost route when the complete relevant graph and costs are known.
3. The model-based agent remembers discovered obstacles and hazards.
4. Online planning enables the robot to respond to new information.
5. The robot can compare safer and riskier routes using their accumulated costs.

One important limitation is that ordinary UCS guarantees global optimality when the relevant graph and costs are known. In a partially observable environment, an online agent may need to replan as it discovers new cells; it cannot guarantee the globally optimal route before learning the necessary information.

Conclusion

A model-based, goal-based online search agent using UCS provides an effective approach for autonomous rescue operations. It updates its knowledge dynamically, avoids obstacles, considers risk costs, and selects low-cost routes to reach survivors safely.

Code of Conduct:

I Madhan R\192412370 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Madhan R

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Problem Understanding	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	6
Constraint Analysis	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	6
Solution Development	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	7.5
Application of Concepts	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	6
Justification	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	4.5